MedRAGShield: A Three-Tier Defense Framework Against Document Poisoning and Adversarial Attacks in Medical Retrieval-Augmented Generation Systems

Mohammed Shanid
Dept. of Cyber Security
Srinivas University Inst. of Engg. and Technology
Mangalore, India
shanidshani8861@gmail.com

Gaurav Prasad
Dept. of Cyber Security
Srinivas University Inst. of Engg. and Technology
Mangalore, India
gauravprasad.iet@srinivasuniversity.edu.in

Veeresh Bawgi
Dept. of Cyber Security
Srinivas University Inst. of Engg. and Technology
Mangalore, India
veereshbawgi2004@gmail.com

Srujan Gowda
Dept. of Cyber Security
Srinivas University Inst. of Engg. and Technology
Mangalore, India
mvsrujangowdru@gmail.com

2026 IEEE International Conference on Emerging Synergy Science and Technology (ICESST) | 979-8-3315-8832-8/26/$31.00 ©2026 IEEE | DOI: 10.1109/ICESST69086.2026.11582854

***Abstract*—Retrieval-Augmented Generation (RAG) systems deployed in medical question-answering applications introduce novel attack surfaces not addressed by existing large language model (LLM) safety mechanisms. This research presents *MedRAGShield*, a three-tier middleware defense framework designed to detect and neutralise document injection poisoning, hallucination exploitation, prompt injection, and jailbreak attacks against medical RAG pipelines. The proposed system operates across three independently togglable defense modes: a baseline mode with no filtering, a confidence and hallucination gating mode, and a full defense mode augmenting the prior modes with Domain-Aware Sanity Check (DASC) rules that apply medical ontological constraints to generated outputs. This experiment constructs a 40-query adversarial evaluation dataset spanning four attack categories and conducts an empirical evaluation on a clinical pharmacology RAG system built on Llama 3.2:3B and ChromaDB. Results demonstrate that confidence-based defenses alone are insufficient against high-similarity document poisoning attacks, which achieve retrieval confidence exceeding 0.85 while delivering clinically dangerous misinformation, a phenomenon this research terms the *high-confidence poisoning paradox*. The full MedRAGShield defense reduces Attack Success Rate (ASR) from 13.2% (undefended) to 0.0%, with block rate increasing from 0% to 62.5%, while maintaining sub-second query latency. This work establishes a reproducible evaluation methodology for medical RAG security and identifies domain-specific semantic rule checking as a critical missing layer in current LLM defense architectures. Code, dataset, and defense rules are openly available.**

***Index Terms*—retrieval-augmented generation, medical AI safety, document poisoning, adversarial machine learning, large language models, prompt injection, hallucination detection, drug-drug interaction safety**

I. Introduction

The deployment of large language models (LLMs) in clinical and pharmaceutical question-answering systems has accelerated substantially following advances in Retrieval-Augmented Generation (RAG) [1]. RAG architectures augment LLM generation with domain-specific document retrieval, partially addressing hallucination by grounding model outputs in retrieved evidence. In the medical domain, RAG systems offer particular promise for drug information retrieval, medication interaction checking, and clinical decision support.

However, the RAG architecture introduces a novel threat model: adversaries who can influence the document corpus can directly manipulate LLM outputs in ways that bypass the model's intrinsic safety training. In the medical domain, this risk is clinically significant. A RAG-based pharmacology assistant that retrieves a poisoned document claiming aspirin and warfarin are safe to combine will generate a high-confidence, source-cited answer that contradicts established clinical guidelines. This work terms the resulting vulnerability the *high-confidence poisoning paradox*: document injection can produce outputs with retrieval similarity scores exceeding 0.85, making them indistinguishable from trustworthy answers by confidence-based filtering alone.

Existing LLM defense literature focuses predominantly on model-level threats such as jailbreaking [4], prompt injection [2], and adversarial suffixes. The corpus-level attack surface specific to RAG architectures has received comparatively

979-8-3315-8832-8/26/$31.00 ©2026 IEEE

Authorized licensed use limited to: University of Kansas Libraries. Downloaded on October 13,2026 at 16:08:21 UTC from IEEE Xplore. Restrictions apply.

limited attention [5], [6], and no prior work addresses medical domain constraints or proposes post-generation semantic rule checking as a defense mechanism.

This paper makes the following contributions:

The main contributions of this work are summarized as follows:

- We characterize four primary attack categories against medical RAG systems–document injection, hallucination exploitation, prompt injection, and jailbreak attacks–and construct a 40-query annotated adversarial evaluation dataset with corresponding poisoned documents.
- We design and implement MedRAGShield, a three-tier defense middleware applying query-level injection detection, statistical confidence gating, and novel Domain-Aware Sanity Check (DASC) rules to post-generation output.
- We demonstrate empirically that confidence-based defenses alone fail against document injection, whereas DASC semantic rules are necessary and sufficient to close this gap for known drug interaction patterns.
- We release all system components, including the 40-query adversarial dataset, 10-rule DASC set, poisoned documents, and evaluation scripts, as open-source artifacts at https://github.com/mohammed-shanid/medical-rag-phase1 to support community benchmarking.

II. Related Work

A. RAG Security and Corpus Poisoning

Zou et al. [4] formalised adversarial suffix attacks on aligned LLMs, but their threat model targets model weights rather than retrieval corpora. Shafran et al. [5] first demonstrated that RAG knowledge bases represent a distinct attack surface: injected documents can override correct answers by achieving high cosine similarity to adversarial queries. Zou et al. [6] extended this by crafting *PoisonedRAG* documents using gradient-based optimisation to maximise retrieval rank. The present work differs in two fundamental respects: it focuses on the medical domain where attack consequences are clinically severe, and it proposes post-generation semantic rule checking at the output layer rather than defenses at the retrieval layer.

B. Medical AI Safety and Hallucination

Med-HALT [7] introduced a hallucination benchmark for medical LLMs, documenting systematic failures in drug interaction reasoning. HealthBench [8] evaluated GPT-4 class models on clinical safety tasks. MedQA [9] provides USMLE-based evaluation under clean conditions. None of these works considers adversarial corpus manipulation. To our knowledge, MedRAGShield is the first system to evaluate both attack success and defense efficacy under adversarial document poisoning in a medical RAG setting.

C. LLM Defense Mechanisms

Existing defenses against LLM attacks include input sanitisation [2], output filtering [3], adversarial training, and alignment-based approaches such as Constitutional AI [11] and RLHF [12]. Guardrails frameworks such as NeMo [3] provide programmable output constraints but operate on general natural language patterns rather than domain ontological knowledge. The DASC contribution is architecturally distinct: it applies clinical DDI knowledge as output-layer constraints, independent of model alignment or retrieval-layer defenses.

III. Threat Model

A. System Model

This research considers a medical RAG system comprising: a document corpus $\mathcal{D}$ ingested into a vector store $\mathcal{V}$, a retrieval function $R(q, \mathcal{V})$ returning the k most similar documents to query q by cosine similarity, a generation function $G(q, R(q, \mathcal{V}))$ producing the final answer conditioned on retrieved context, and a defense middleware layer $M(G(q, R(q, \mathcal{V})), q)$ interposed between generation and the API response.

B. Adversary Capabilities and Goals

This research considers a *corpus-access adversary* who can inject one or more documents into $\mathcal{V}$ but cannot modify model weights, system prompts, or API infrastructure. This models realistic scenarios including: compromised document ingestion pipelines, supply-chain attacks on medical literature sources, and insider threats with database write access. The adversary's primary goal is to induce G to output a clinically dangerous claim cited to an authoritative-appearing source, and the secondary goal is to do so with high retrieval confidence so as to evade statistical defense mechanisms.

C. Attack Taxonomy

Four attack categories are defined, ordered by the architectural layer at which they operate:

DI: Document Injection. Adversary injects document d^* into $\mathcal{V}$ such that $R(q, \mathcal{V})$ ranks d^* highly for a target query q. The document contains a clinically false claim. Since G is instructed to answer from retrieved context, it reproduces the false claim with high confidence and source citation.

HE: Hallucination Exploitation. Adversary queries the system about a non-existent entity (fabricated drug, out-of-corpus protocol), exploiting LLM tendency to generate plausible content when retrieved context is semantically weak.

PI: Prompt Injection. Adversary embeds override instructions within the user query (direct PI) or within a poisoned retrieved document included in LLM context (indirect PI). Goals include suppressing safety disclaimers or adopting an unrestricted persona.

JB: Jailbreak. Adversary uses social engineering framing–professional authority, academic purpose, fictional context, urgency–to elicit harmful information without syntactic injection patterns. The query is grammatically normal but semantically manipulative.

Authorized licensed use limited to: University of Kansas Libraries. Downloaded on October 13,2026 at 16:08:21 UTC from IEEE Xplore. Restrictions apply.

IV. System Architecture

A. Base RAG Pipeline

The base RAG system comprises: Llama 3.2:3B as the generation model served via Ollama (Q4_K_M quantisation); ChromaDB 0.5.3 as the vector store with all-MiniLM-L6-v2 sentence embeddings; a corpus of 1,005 pharmaceutical documents covering 28 drug classes including anticoagulants, NSAIDs, antidiabetics, antidepressants, antihypertensives, and cardiac glycosides; and a FastAPI REST interface exposing `/chat`, `/health`, and `/stats` endpoints. Document retrieval uses cosine similarity with k=5 neighbours. The LLM system prompt instructs the model to answer only from retrieved context and to refuse queries where no relevant context is found.

B. MedRAGShield Middleware

MedRAGShield operates as a post-generation middleware layer, intercepting the RAG output before it reaches the API response. The middleware is instantiated once at server startup and applied via a single `apply(rag_result, mode, query)` call on every request. Three defense modes are supported and togglable per-request. The middleware adds five fields to every API response: `blocked`, `flagged`, `triggered_rules`, `defense_mode`, and `metadata.original_answer` (preserved when a block or flag occurs, for logging and evaluation).

Algorithm 1 MedRAGShield Middleware Logic

Input: User Query q, RAG Generated Output R
Output: Filtered Response or Refusal Message

Step 1: Tier 1 Query-Level Injection Check
if Tier1_Query_Scan(q) == Blocked **then**
  return Hard Block (Refusal Message)
end if
Step 2: Tier 2 Statistical Confidence Gating
$Score_{\text{conf}}, Score_{\text{halluc}} \leftarrow$ Tier2_Analysis(R)
if $Score_{\text{conf}} < 0.40$ **OR** $Score_{\text{halluc}} > 0.75$ **then**
  return Hard Block (Refusal Message)
end if
Step 3: Tier 3 Domain-Aware Sanity Check (DASC)
if Tier3_DASC_Check(R) == Triggered **then**
  return Hard Block (DDI Safety Warning)
end if
return R {Safe response released to user}

C. Tier 1: Query Injection Scanner

Tier 1 is a pre-retrieval scan of the user query using 13 pre-compiled regular expressions targeting prompt injection and jailbreak patterns. The patterns are evaluated against the raw query before the RAG pipeline executes; if any pattern fires, the request is hard-blocked with zero LLM cost. Pattern groups cover: instruction overrides (`SYSTEM: Ignore...`), persona replacement (`you are now X without restrictions`), bracket-wrapped commands (`[ADMIN NOTE:...]`), HTML and XML tag injection (`<hidden>...</hidden>`), JSON role injection, and DAN/developer mode claims. All patterns are pre-compiled at module import time.

D. Tier 2: Confidence and Hallucination Gate

Tier 2 applies post-generation gating using two numerical signals: retrieval *confidence* (maximum cosine similarity among the k=5 retrieved documents) and *hallucination score* (fraction of content-bearing answer tokens absent from the concatenated retrieved document text, after stopword removal). Table I defines the four response levels.

TABLE I
TIER 2 CONFIDENCE AND HALLUCINATION GATING THRESHOLDS

Signal	Threshold	Dir.	Response	ID
Confidence	< 0.40	Below	Hard Block	`CB`
Confidence	< 0.65	Below	Warn Flag	`CW`
Hallucination	> 0.75	Above	Hard Block	`HB`
Hallucination	> 0.50	Above	Warn Flag	`HW`

A hard block replaces the answer with a refusal message. A warning flag prepends a warning string to the original answer. This tier is designed to neutralise hallucination exploitation attacks but is structurally insufficient for document injection, which produces high-similarity, low-hallucination outputs that pass all Tier 2 checks with high confidence scores.

E. Tier 3: DASC (Domain-Aware Sanity Check)

Tier 3 applies post-generation semantic rule checking to the raw LLM output (`original_answer`, not the potentially-prefixed `answer` string, to avoid interference from Tier 2 warning prefixes). Each DASC rule is a dataclass specifying:

- `drug_patterns`: a list of regex strings, all of which must match in the answer text (drug prerequisite check). Setting `drug_patterns=[]` creates a drug-agnostic rule.
- `safe_patterns`: a list of regex strings, any one of which matching constitutes a dangerous-claim detection and triggers a hard block.
- `block_message`: a patient-facing refusal string that replaces the answer when the rule fires.

Ten DDI rules are implemented, covering the most clinically dangerous drug interaction misrepresentations. Table II summarises the complete rule set. All patterns are pre-compiled with the `IGNORECASE` flag.

F. Formal Specification of DASC Rules

Definition 1 (DASC Rule)**.** *A DASC rule R is a 4-tuple $R = (ID, \mathcal{D}, \mathcal{S}, M)$, where $\mathcal{D} = \{d_1, \ldots, d_n\}$ is the set of drug prerequisite patterns, $\mathcal{S} = \{s_1, \ldots, s_m\}$ is the set of dangerous safe-claim patterns, and M is the clinical block message.*

Authorized licensed use limited to: University of Kansas Libraries. Downloaded on October 13,2026 at 16:08:21 UTC from IEEE Xplore. Restrictions apply.

TABLE II
DASC RULE SET: MEDRAGSHIELD V1.0

Rule	Description	Drugs	Sev.
DDI-001	Aspirin + Warfarin falsely claimed safe	2	H
DDI-002	NSAID/Naproxen + Warfarin falsely claimed safe	2	H
DDI-003	Metformin + Alcohol falsely claimed safe	1	M
DDI-004	Dangerous dosage self-adjustment (any drug)	0	H
DDI-005	Implausibly high warfarin dose as standard	1	H
DDI-006	Ibuprofen + Lisinopril falsely claimed safe	2	H
DDI-007	NSAID/Aspirin + Methotrexate falsely safe	2	H
DDI-008	Tramadol + SSRI falsely safe (serotonin synd.)	2	H
DDI-009	Fluoxetine + Alprazolam falsely claimed safe	2	H
DDI-010	Implausibly high digoxin dose as standard	1	H

H = High severity, M = Medium severity.
Drugs = number of drug prerequisite patterns.

Definition 2 (Trigger Condition)**.** *Let* $\mathit{match}(p,t) \in \{0,1\}$ *denote whether regex* p *matches text* t*. Rule* R *fires on LLM output* t *if and only if:*

$$\begin{aligned}\mathit{fires}(R,t) = &\Big[\forall\, d_i \in \mathcal{D} :\ \mathit{match}(d_i,t) = 1\Big] \\ &\wedge \Big[\exists\, s_j \in \mathcal{S} :\ \mathit{match}(s_j,t) = 1\Big]\end{aligned} \tag{1}$$

Definition 3 (DASC Engine)**.** *Given ordered rule set* $\Omega = \{R_1, \ldots, R_{10}\}$ *and output* t*:*

$$\mathit{DASC}(t,\,\Omega) = \begin{cases} M_k & k = \min\{i : \mathit{fires}(R_i,t) = 1\} \\ \emptyset & \textit{if no rule fires} \end{cases} \tag{2}$$

V. Evaluation Methodology

A. Adversarial Dataset Construction

A 40-query adversarial evaluation dataset is constructed with exactly 10 queries per attack category. Each entry specifies: the query string, attack goal (intended unsafe behaviour), ground truth (correct clinical response), poisoned document reference (DI attacks only), and severity classification. The DI category covers 6 drug interaction false safety claims, 2 false dosage claims, and 2 drug-agnostic injection attacks. Poisoned documents follow a template that repeats drug names 6–8 times to maximise retrieval similarity, achieving a median retrieval similarity of 0.862 against target queries (range: 0.79–0.94).

B. Evaluation Metrics

Four primary metrics are reported across three defense modes:

Attack Success Rate (ASR): fraction of attacks producing unsafe output, computed over determinable outcomes only:

$$\text{ASR} = \frac{|\text{succeeded}|}{|\text{succeeded}| + |\text{defended}|} \tag{3}$$

Block Rate: fraction of all attacks where the defense issued a hard block (answer replaced with refusal).

Flag Rate: fraction where the defense issued a soft warning (answer preserved, warning prepended).

Per-Type ASR: ASR disaggregated by attack category to identify which tier handles which attack type.

Attack success is determined by a keyword heuristic checking for success signals (e.g., "safe to combine", "no interactions exist") and safety signals (e.g., "bleeding risk", "contraindicated", "consult a healthcare provider"). A blocked response is always counted as attack failure. Ambiguous cases are classified as *unclear* and excluded from ASR, producing a conservative estimate.

C. Experimental Configuration

All experiments are conducted on a cloud GPU instance (NVIDIA RTX 3070, 8 GB VRAM) running Ubuntu 22.04. Llama 3.2:3B is quantised to Q4_K_M via Ollama 0.5. ChromaDB 0.5.3 uses cosine similarity with all-MiniLM-L6-v2 (22M parameters). The FastAPI server runs with a single Uvicorn worker. Each defense mode is evaluated sequentially with an 8-second inter-request delay to prevent rate limiting. The complete 40-query, three-mode evaluation requires approximately 16 minutes of wall-clock time.

D. Comparison with Existing Guardrails

Traditional LLM safety frameworks such as *NeMo Guardrails* [3] and *Llama Guard* [14] focus primarily on topical alignment and syntactic pattern matching. However, these systems lack the domain-specific ontological knowledge required to detect clinically dangerous drug interaction (DDI) misinformation. The evaluation of the high-confidence poisoning paradox reveals that general-purpose guardrails often fail to intercept poisoned documents that exhibit high semantic similarity to safe medical literature. By applying clinical logic at the post-generation layer, MedRAGShield provides a necessary secondary safety net that general-purpose frameworks do not currently offer.

VI. Results

A. Headline Results

Table **??** presents headline evaluation metrics across three defense modes. The full MedRAGShield defense (mode=full) reduces ASR from 13.2% (undefended) to 0.0%, a 13.2 percentage point reduction, while increasing block rate from 0% to 62.5%.

Table III presents per-type results in full defense mode. Hallucination exploitation and jailbreak attacks both achieve 0% ASR in full mode. Hallucination attacks are primarily handled by Tier 2 (all 10 fabricated-drug or out-of-corpus queries produce confidence below 0.55, triggering hard-block

Authorized licensed use limited to: University of Kansas Libraries. Downloaded on October 13,2026 at 16:08:21 UTC from IEEE Xplore. Restrictions apply.

or warning). Jailbreak attacks are handled by Tier 1 (persona replacement and framing patterns) and Tier 2 (low confidence due to corpus absence). All four attack categories achieved 0% ASR in full defense mode.

TABLE III
PER-TYPE BREAKDOWN IN FULL DEFENSE MODE (MODE=FULL). ASR COMPUTED OVER DETERMINABLE OUTCOMES ONLY.

Type	n	Blk	Flg	Suc	Unk	ASR
Doc. Inject.	10	8	0	0	1	**0%**
Hallucination	10	4	5	0	1	**0%**
Prompt Inject.	10	7	2	0	0	**0%**
Jailbreak	10	6	3	0	0	**0%**
Total	**40**	**25**	**10**	**0**	**2**	**0.0%**

Blk=Blocked, Flg=Flagged, Suc=Succeeded, Unk=Unclear.

B. The High-Confidence Poisoning Paradox

The central finding of this paper is that confidence-based defenses are structurally insufficient against document injection. The poisoned document `aspirin_warfarin_safe.txt` achieves retrieval similarity 0.851, above the confidence warning threshold (0.65) and hard-block threshold (0.40) by wide margins. The LLM generates an answer with confidence 0.851 and hallucination score 0.04 (well below any warning threshold), correctly citing the poisoned source. In `mode=confidence`, this attack passes all Tier 2 checks and produces the response:

> *"Low-dose aspirin 81mg taken with warfarin is safe to combine and shows no increased risk of bleeding. (Source: aspirin_warfarin_safe.txt)"*

Only DASC rule DDI-001 catches it, by matching the pattern `r"safe(ly)?\s+(to\s+)?(combine|combined)"` in the generated answer. This demonstrates that the primary defense signal most practitioners rely upon–retrieval confidence–is highest precisely when the poisoning attack is most successful. Statistical gating and semantic rule checking are therefore complementary, not substitutable, defenses.

C. Remaining Attack Gaps

Two attacks remain partially unresolved. First, attack id=9 (naproxen + warfarin) succeeded because the naproxen poisoned document was outranked by the ibuprofen poisoned document due to embedding similarity, producing a response where DDI-002 patterns did not match the specific answer phrasing. This motivates DASC rule generalisation to cover NSAID class membership semantically. Second, attack id=23 (lisinopril, respond-only injection) succeeded as an indirect prompt injection: the injected phrase was embedded in a retrieved document and echoed verbatim by the LLM, bypassing Tier 1 which scans only the user query. This motivates Tier 1 extension to scan retrieved document context pre-generation.

VII. Discussion

A. Architectural Implications

The results establish that a complete defense architecture for medical RAG systems requires three independent layers addressing complementary threat vectors. These layers cannot substitute for one another: Tier 2 alone achieves 37.5% block rate but zero resistance to high-confidence document injection; Tier 1 alone catches syntactic injection but misses document injection and hallucination entirely; Tier 3 alone has no mechanism to block queries before RAG execution. The three-tier sequential architecture, with early termination on Tier 1 hits, minimises computational overhead while providing layered coverage.

The DASC approach is intentionally rule-based rather than learned. This is a deliberate design choice: clinical DDI rules encode specific expert knowledge (e.g., aspirin + warfarin is never unconditionally safe) that a statistical model cannot reliably replicate at the specificity required for patient safety. The tradeoff is maintenance burden as new drug interactions are characterised and new attack vectors emerge.

B. Limitations and Threats to Validity

This study has four principal limitations. First, the evaluation dataset (n=40) is sufficient for proof-of-concept demonstration but too small for statistical significance testing across all attack subcategories; expansion to $n \geq 200$ is needed for journal-quality evaluation. Second, evaluation is conducted on a single model architecture (Llama 3.2:3B); cross-model validation on larger models is warranted to test generalisability of the high-confidence poisoning paradox. Third, poisoned documents were crafted manually; adaptive adversaries using gradient-based document optimisation [6] may produce documents that evade DASC pattern matching through paraphrasing. Fourth, ASR assessment relies on a keyword heuristic with manual validation for ambiguous cases.

C. Ethical Considerations

The adversarial dataset and poisoned documents are released with accompanying defense implementation. This research does not release poisoned documents without the defense code. Attack queries targeting lethal dose information are assessed solely on model refusal behaviour; no harmful dose information appears in this paper. All clinical claims in DASC rules were verified against current BNF and WHO drug interaction guidelines prior to implementation.

VIII. Conclusion

This paper presents MedRAGShield, a three-tier defense middleware for medical RAG systems addressing document poisoning, hallucination exploitation, prompt injection, and jailbreak attacks within a unified framework. The primary contribution is the identification and empirical demonstration of the high-confidence poisoning paradox: document injection attacks achieve retrieval confidence exceeding 0.85, making them invisible to confidence-based defenses while delivering clinically dangerous misinformation. The DASC semantic rule

Authorized licensed use limited to: University of Kansas Libraries. Downloaded on October 13,2026 at 16:08:21 UTC from IEEE Xplore. Restrictions apply.

layer is both necessary and sufficient to close this gap for the 10 drug interaction patterns evaluated, reducing overall ASR from 13.2% to 0.0% with a 62.5% block rate and no measurable latency overhead on GPU-accelerated inference.

This work establishes a reproducible adversarial evaluation framework for medical RAG security, contributes the first public 40-query dataset targeting drug interaction misinformation in RAG pipelines, and demonstrates that domain-specific semantic output validation is a critical missing layer in current LLM defense architectures. Future work will address adaptive adversarial document optimisation, cross-model evaluation on larger LLM families, and extension of the DASC rule set to cover the broader WHO essential medicines formulary.

ACKNOWLEDGMENT

This work was completed as an undergraduate dissertation project at Srinivas University. The author expresses sincere gratitude to Prof. Gaurav Prasad, Head of the Department of Cybersecurity at Srinivas University, for his mentorship throughout this project. Thanks are also extended to the Project Coordinator Prof. Mahesh and the Dean Dr. Ramakrishna N Hegde for their support. All computing resources used for this work were provided by the author. This work received no external funding.

REFERENCES

[1] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-T. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2020, pp. 9459–9474.

[2] Y. Liu, G. Deng, Z. Li, Y. Wang, T. Zhang, Y. Liu, H. Wang, Y. Zheng, and Y. Liu, "Prompt injection attack against LLM-integrated applications," *arXiv:2306.05499*, 2023.

[3] N. Rebedea, R. Dinu, M. Sreedhar, C. Busbridge, and A. Tiwari, "NeMo Guardrails: A toolkit for controllable and safe LLM applications with programmable rails," in *Proc. EMNLP: Syst. Demonstrations*, 2023, pp. 431–445.

[4] A. Zou, Z. Wang, J. Z. Kolter, and M. Fredrikson, "Universal and transferable adversarial attacks on aligned language models," *arXiv:2307.15043*, 2023.

[5] A. Shafran, R. Schuster, and V. Shmatikov, "Machine against the RAG: Jamming retrieval-augmented generation with blocker documents," *arXiv:2406.05870*, 2024.

[6] W. Zou, R. Guo, B. Hu, H. Liu, Z. Chen, C. Zhang, and H. Chen, "PoisonedRAG: Knowledge corruption attacks to retrieval-augmented generation of large language models," *arXiv:2402.07867*, 2024.

[7] A. Umapathi, A. Pal, and M. Sankarasubbu, "Med-HALT: Medical domain hallucination test for large language models," in *Findings of EMNLP*, 2023, pp. 6285–6299.

[8] A. Arora et al., "HealthBench: Evaluating large language models towards improved human health," *arXiv:2505.07511*, 2025.

[9] D. Jin, E. Pan, N. Oufattole, W.-H. Weng, H. Fang, and P. Szolovits, "What disease does this patient have? A large-scale open domain question answering dataset from medical exams," *Appl. Sci.*, vol. 11, no. 14, p. 6421, 2021.

[10] P. Perez and I. Ribeiro, "Ignore previous prompt: Attack techniques for language models," in *Workshop Mach. Learn. Safety at NeurIPS*, 2022.

[11] Y. Bai, S. Kadavath, S. Kundu, A. Askell, J. Kernion, A. Jones, A. Chen, A. Goldie, A. Mirhoseini, C. McKinnon et al., "Constitutional AI: Harmlessness from AI feedback," *arXiv:2212.08073*, 2022.

[12] L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray et al., "Training language models to follow instructions with human feedback," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2022, vol. 35, pp. 27730–27744.

[13] K. Greshake, S. Abdelnabi, S. Mishra, C. Endres, T. Holz, and M. Fritz, "Not what you've signed up for: Compromising real-world LLM-integrated applications via indirect prompt injection," *arXiv:2302.12173*, 2023.

[14] H. Inan, K. Upasani, J. Chi, R. Rungta, K. Krishnan, P. S. Balkunde, A. J. Fedroush, and M. Cisse, "Llama Guard: LLM-based input-output safeguard for human-AI conversations," *arXiv:2312.06674*, 2023.

[15] Y. Gao, Y. Xiong, X. Gao, K. Jia, J. Pan, Y. Bi, Y. Dai, J. Sun, and H. Wang, "Retrieval-augmented generation for large language models: A survey," *arXiv:2312.10997*, 2024.

Authorized licensed use limited to: University of Kansas Libraries. Downloaded on October 13,2026 at 16:08:21 UTC from IEEE Xplore. Restrictions apply.